Mohamad Hosein Bemani

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Computer Engineering student interested in systems programming, compilers and concurrency, with internship experience in ASP.NET Core and Angular. Seeking internship and junior software engineering roles (backend, full-stack, systems).

EXPERIENCE

Software Developer Intern, National Iranian Copper Industries Co. (Sarcheshmeh Copper Complex), IT unit Aug 2026 - Sep 2026

- Studied an enterprise overtime-distribution system built with Clean Architecture, DDD and CQRS (MediatR).
- Completed a delegation feature after checking every backend and frontend consumer; delivered as Git patch files.
- Added server-side sorting, filtering and search to the main tables (PrimeNG) and wrote guidance for other modules.
- On an ID-card system (ASP.NET Core, .NET 8): analysed a script engine that recompiled on every request and built a compile cache.

EDUCATION

B.Sc. Computer Engineering, Shahid Beheshti University Sep 2022 - expected 2027

Operating Systems, Compilers, Computer Networks, Cryptography, AI, Databases, Digital System Design, Embedded Systems, Machine Learning, Software Engineering

PROJECTS

- **AI-Automation-Studio** (Python, TypeScript): local pipeline from storyboard to narrated 1080p and vertical videos (Kokoro, Whisper, Remotion, FFmpeg, n8n); v1.0.0 with a quickstart tested from a fresh clone.
- **Multithreaded message broker** (C, pthreads): priority queues, TTL, deadline-aware scheduling; verified with ThreadSanitizer in CI.
- **Toy language compiler** (C++17, LLVM): lexer, recursive-descent parser, optimizer and LLVM IR code generation; CI builds it and runs a sample.
- **Block cipher and certificate authority** (Java): 128-bit cipher with CBC/OFB/CTR modes, RSA/DSA and a toy CA (educational).
- **Tanks 2D** (Unity, C#): turn-based artillery game, about 2,500 lines of C#. **Hotel reservation server** (Python sockets) and **Pong** (8086 assembly).

SKILLS

Languages: C, C++, C#, Java, Python, TypeScript, x86 assembly. **Backend:** ASP.NET Core, Entity Framework Core, SQL Server, Angular. **Systems:** concurrency, operating systems, networking, cryptography, compilers, LLVM. **Tools:** Git, Docker, GitHub Actions, FFmpeg, Unity, Blender.